

DFMEA - t7_r2 (screening-level)

VBF-T7R2-DFMEA-001

1. Li plating during 4C charge -> controls: fine anode 2.0 μm + t+ 0.5 + anode porosity 0.32; residual risk: +9 mV margin is SoC-dependent.
2. Nail thermal runaway -> control: $h=80 \text{ W/m}^2/\text{K}$ cooling equilibrates 10 W source at 321.73 K, below the ~356 K ignition observed in the R2 hot probe.
3. SEI overgrowth @45 C -> control: 463.1 nm vs 550 nm limit (87 nm margin).
4. ED shortfall -> control: 423.6 vs 327.2 (+96 Wh/kg); electrolyte excluded by contract.
5. High-porosity separator path -> avoided: F4 (sep 0.55) not selected.